PULLBACK ALONG A MORPHISM ADMITTING A MAP BACK OVER THE BASE IS AN EQUIVALENCE ON DESCENT DATA

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 2 navigation stubs; the reasoning that turns the cited self-map statement into an equivalence is displayed in the span.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Editorial note. The canonical tag of this label does not occur among the tags of the published items of descent.tex, so it is not a published duplicate.

Allowed standard prerequisites (editorial).

- Pullback along a morphism from a scheme to itself over the base is isomorphic to the identity. Same chapter, Lemma lemma-pullback-selfmap, invoked at proof lines 8908-8910. Supplies the identification of both composites with the identity functors.
- The pullback functor on descent data attached to a morphism over a base. Same chapter, Lemma lemma-pullback, named in the statement at lines 8900-8901. Supplies the functor whose being an equivalence is proved.

Source reference. Original lemma 023Y: sources/stacks-descent/pinned-original-source.tex, source lines 8496–8496. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0242: sources/stacks-descent/pinned-original-source.tex, source lines 8883–8883. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X' \to X$ be a morphism of schemes over a base scheme S. Assume there exists a morphism $g : X \to X'$ over S, for example if f has a section. Then the pullback functor of Lemma 023Y defines an equivalence of categories between the category of descent data relative to X/S and X'/S.*

Proof. Let $g : X \to X'$ be a morphism over S. Lemma 0242 above shows that the functors $f^* \circ g^* = (g \circ f)^*$ and $g^* \circ f^* = (f \circ g)^*$ are isomorphic to the respective identity functors as desired. □

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.